Code quality
Laborator 6

Tehnici avansate de programare

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Code quality metrics

- Software Improvement Group - Maintainability: guidance for producers

- https://softwareimprovementgroup.com/wp-content/uploads/SIG-TUViT-Evaluation-Criteria-Trusted-Product-Maintainability-Guidance-for-producers.pdf

Establish teams

- Teams of 4-5 students
  - Total: 33 studenti

- Team name

- Team owner – email for communication

Team owner responsibilities

- Provide board link to teacher
- Ensure product backlog split by work items
- Monitor product delivery
- Ensure all blockers are solved in time
- Can communicate with the teacher between courses for unblocking issues

Project assignments

- Project assignment.pdf

Team decisions

- Team names and member structure
- Team owner
- Short presentation per team member
- Project

GIT branch strategy

- New Azure DevOps repository
  - Can be accessed by team members and teacher
  - .gitignore for VisualStudio
- Main branch – stable for all team members
- Each team member creates from main a new branch for new task changes
- The task branch will be merged to main through a PullRequest with a mandatory code reviewer

GIT changes

- Create branch from command line/Visual Studio

- Create commit from Visual Studio
  - Git changes
  - Staged changes
  - Meaningful commit description
  - Stash

GIT PullRequest

- Create PullRequest from browser
  - Choose source and destination branch for the PullRequest
  - Add reviewer
  - Add work item
  - Add comments

- GIT parallel work
  - Merge from main into the task branch

PullRequest code review checklist

- Business requirements
- Code quality
- Security
- Architecture

Project board planning

- https://azure.microsoft.com/en-us/products/devops

Videos

- Mastering Unit Tests — First Look at The Art of Unit Testing, Third Edition

- https://learn.microsoft.com/en-us/shows/visual-studio-toolbox/modern-dev-practices-unit-testing

Learning sources

- https://learn.microsoft.com/en-us/dotnet/csharp/fundamentals/coding-style/identifier-names
- https://learn.microsoft.com/en-us/dotnet/csharp/fundamentals/coding-style/coding-conventions
- https://enterprisecraftsmanship.com/posts/3-things-that-will-make-or-break-your-project/
- https://enterprisecraftsmanship.com/posts/exceptions-for-flow-control/

Learning sources

- https://enterprisecraftsmanship.com/posts/error-handling-exception-or-result/
- https://enterprisecraftsmanship.com/posts/functional-c-handling-failures-input-errors/
- https://enterprisecraftsmanship.com/posts/pragmatic-unit-testing/
- https://enterprisecraftsmanship.com/posts/pragmatic-integration-testing/
- https://learn.microsoft.com/en-us/dotnet/core/testing/

Learning sources

- https://docs.shouldly.org/
- https://xunit.net/docs/getting-started/v3/getting-started
- https://fakeiteasy.github.io/
- https://code.visualstudio.com/docs/csharp/testing
- https://learn.microsoft.com/en-us/visualstudio/test/walkthrough-creating-and-running-unit-tests-for-managed-code?view=visualstudio
- The Art of Unit Testing with Roy Osherove

Exercises

- https://learn.microsoft.com/en-us/dotnet/core/tutorials/test-class-library?pivots=visualstudio
- https://learn.microsoft.com/en-us/dotnet/core/testing/unit-testing-csharp-with-mstest